

Cadence Generic Process Design Kit: PVS Fill Runset Documentation version 0.5

Section 1: Overview

The Generic Process Design Kit (GPDK) PVS Fill Runset demonstrates common fill use models and features. It supports the following use models:

1. **Command line flow** (batch) – designed for full chip GDSII or OASIS based flows
2. **Large-scale Virtuoso flow using ipvsCreateFill** – designed for larger Virtuoso designs where multiple processors might be helpful
3. **Small-scale Virtuoso flow using the Virtuoso Automated Layout Enhancer (VALE) framework** – designed for performing fill during the design process

The Fill Runset (**version 0.5**) supports the follow features:

1. Virtuoso In-design and batch modes
2. Design type selection
3. Open Access, GDSII and OASIS formats
4. Data extents or prBoundary limited run area
5. Edge halo for spacing the fill inside the boundary by a specified distance
6. Instance-based fill flow
7. Critical net fill flow

Section 2: Run Setup

The GPDK Fill Runset was tested with the following tools and versions:

- ICADV 12.30.720
- PVS 16.12.000
- MVS 18.11.000

The GPDK installs with a number of sub-directories. One of these sub-directories is the “fill” directory.

1. **For the Command line flow:**
 - a. Update the PATH environment variable for Cadence PVS and licensing
 - b. Create a local run directory, and change directories there
 - c. Set the environment variable GPDK_INSTALL_DIR to the parent directory of the fill sub-directory
 - d. Use the example GDSII file midcell_1.gds, and run:
`$GPDK_INSTALL_DIR/fill/ex_runPVSBatchFill_gds.csh`
`$GPDK_INSTALL_DIR/fill/Include/midcell_1.gds midcell_1`
 - e. Examine output midcell_1_merged.gds

2. **For the Large-scale Virtuoso flow using ipvsCreateFill:**
 - a. Set environment variables for Cadence PVS and licensing, in addition to your current Virtuoso setup.
 - b. Add the following lines to your .cdsinit file:
 - i. **NOTE 1:** Fill Runset switches may be set in the custom PVS Fill Form that appears.
 - ii. **NOTE 2:** If the instance fill option is set, *the fill cells must exist in the current library* in order to use them.
 - iii. (if not set) Set environment variable GPDK_INSTALL_DIR:
`setShellEnvVar("GPDK_INSTALL_DIR"
"/path/to/parent/of/fill/directory")`
 - iv. Load the SKIL script:
`load(strcat(getShellEnvVar("GPDK_INSTALL_DIR")
"/fill/runPVSFill.il"))`
 - c. Open a layout window
 - d. Set the runset switches (see section 3 below)
 - e. Use the pull-down menu ipvsCreateFill -> PVS Run Fill
 - f. The results will appear as an instance in the current cellview
3. **For the Small-scale Virtuoso flow using VALE framework:**
 - a. Update the PATH environment variable for Cadence MVS, PVS and licensing, in addition to your current Virtuoso setup.
 - b. NOTE: Given the environment variable MVSHOME is set to the install directory of MVS, it is required to set the environment variable DFMHOME to MVSHOME.
 - c. NOTE: Each of the Runset Options below has a corresponding variable in the VALE Framework configuration file "IMFdefaults.il", only that each variable name has a "IMF_" prefix.
 - d. Add the following lines to your .cdsinit file
 - i. (if not set) Set environment variable GPDK_INSTALL_DIR:
`setShellEnvVar("GPDK_INSTALL_DIR"
"/path/to/parent/of/fill/directory")`
 - ii. Load the SKIL script:
`load(strcat(getShellEnvVar("GPDK_INSTALL_DIR")
"/vale/cdsinitVALE.il"))`
 - e. Open a layout window
 - f. Use the pull-down menu VALE -> Design Finishing -> Run Standard Fill
 - g. The results will appear in the current cellview

Section 3: Runset Options

Environment Variable or PVS switch	Values (default value)	Description
GPDK_FILL_CONFIG	Defined or undefined (undefined)	By default, this switch is not used. It provides a method to configure the Fill runset without environment variables. To use, create a top level PVL file that contains: #DEFINE GPDK_FILL_CONFIG #DEFINE SWITCH1 #UNDEFINE SWITCH2 #INCLUDE /path/to/StdFill*.pvl
DESIGN_TYPE	CELL or CHIP (CELL)	Enables different fill output based on the type of design
BEOL_STACK	8M_7Mx_MT or 10M_9Mx_MT (8M_7Mx_MT)	Sets the metal stack
LAYOUT_SYSTEM	OASIS, GDS, or GDSII (GDSII)	Sets the input data type
LAYOUT_OUTPUT	OASIS, GDS, or GDSII (GDSII)	Sets the output data type
SELECT_PRBOUNDARY	TRUE, YES, FALSE, NO, DEFAULT (YES)	Instructs the runset to use the prBoundary object instead of the extent of the shapes in the design.
SELECT_INST_FILL	TRUE, YES, FALSE, NO, DEFAULT (NO)	Enables instance-based fill
SELECT_CRITICAL_NET_FEATURE	TRUE, YES, FALSE, NO, DEFAULT (NO)	Enables custom spacing around critical nets
CRITICAL_NET_FILE	defined or undefined (undefined)	If defined, it points to a file defining the following variables for the critical net spacings: variable M1_CRITNET_DIST 0.124 variable M2_CRITNET_DIST 0.124 variable M3_CRITNET_DIST

		0.124 variable M4_CRITNET_DIST 0.124 variable M5_CRITNET_DIST 0.148 variable M6_CRITNET_DIST 0.148 variable M7_CRITNET_DIST 0.196 variable MT_CRITNET_DIST 0.384
SELECT_IN_DESIGN_FLOW	TRUE, YES, FALSE, NO, DEFAULT (NO)	When set to TRUE or YES, it will disable the output statements needed for a batch or command line flow.
FILL_EDGE_HALO	float value (0.0)	The boundary is shrunk by the float value amount before generating fill.

Section 4: Licensing Information

In addition to any Virtuoso licenses that might be required, the following PVS and MVS licenses, per flow, are:

1. **Command line**
 - a. 96210 - Physical Verification System Design Rule Checker XL
 - b. 96320 - Physical Verification System Advanced Analysis Option for PVS DRC XL
 - c. 96302 - Cadence Physical Verification System **Advanced Metal Fill** Option
2. **ipvsCreateFill**
 - a. 96310 - Virtuoso IPVS AAO
 - b. 96400 - Virtuoso IPVS Option for Virtuoso
 - c. 73000NG - Physical Verification System Next Generation Product
3. **VALE framework**
 - a. 95311 - Virtuoso DFM, or LPA120 - Litho Hotspot Fixing Option (provides a pack of Virtuoso_VALE_framework license features)
 - b. 96210 - Physical Verification System Design Rule Checker XL
 - c. 96320 - Physical Verification System Advanced Analysis Option for PVS DRC XL
 - d. 96302 - Cadence Physical Verification System **Advanced Metal Fill** Option
 - e. For other VALE Component Modules contained in the GPDK, the following licenses may be required:
 - i. 96340 - Cadence(R) Physical Verification System **Pattern Matching** Option
 - ii. LPA120 - Litho Hotspot Fixing Option